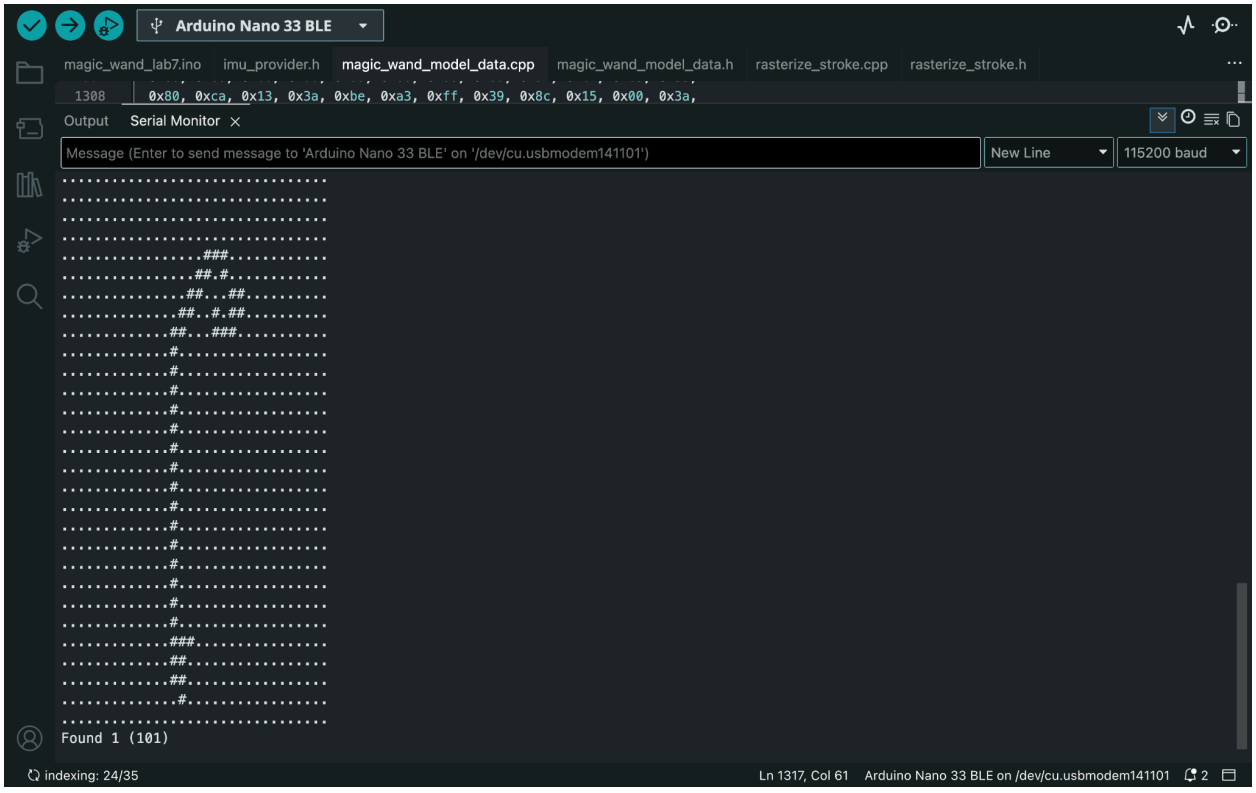
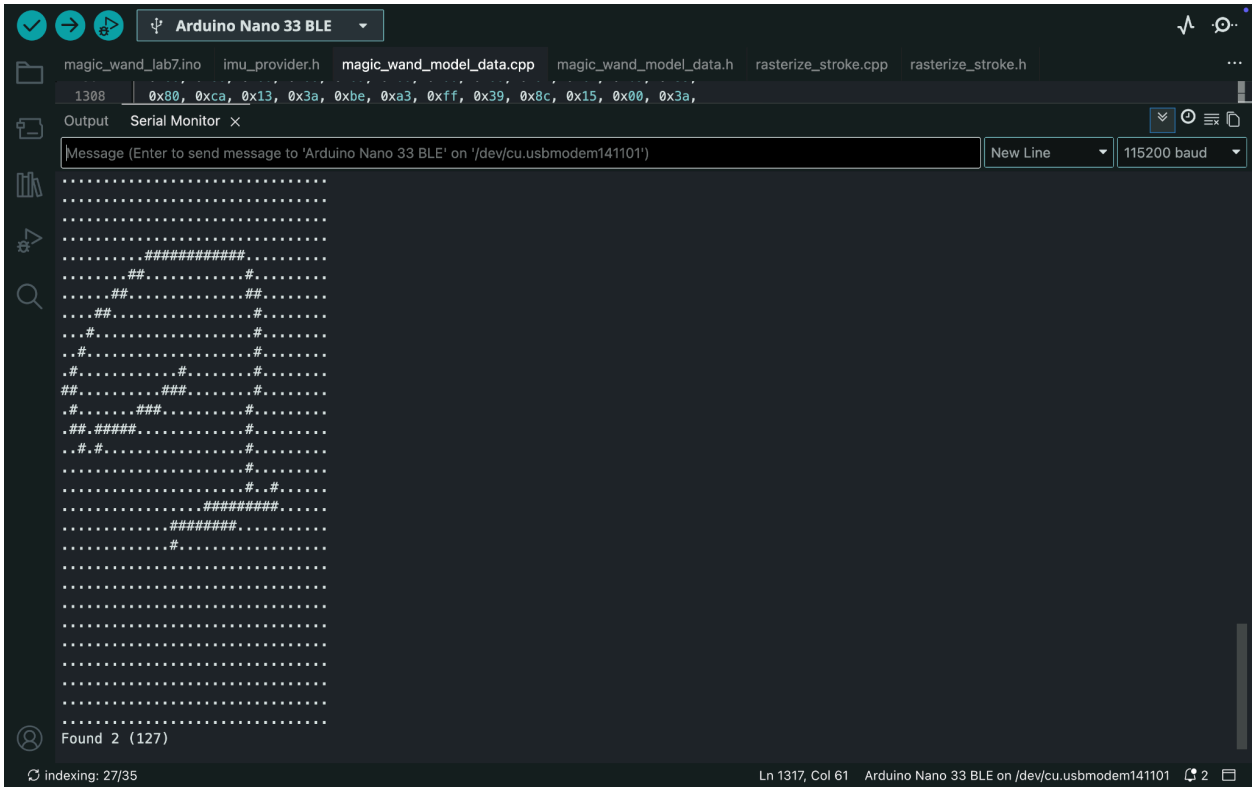
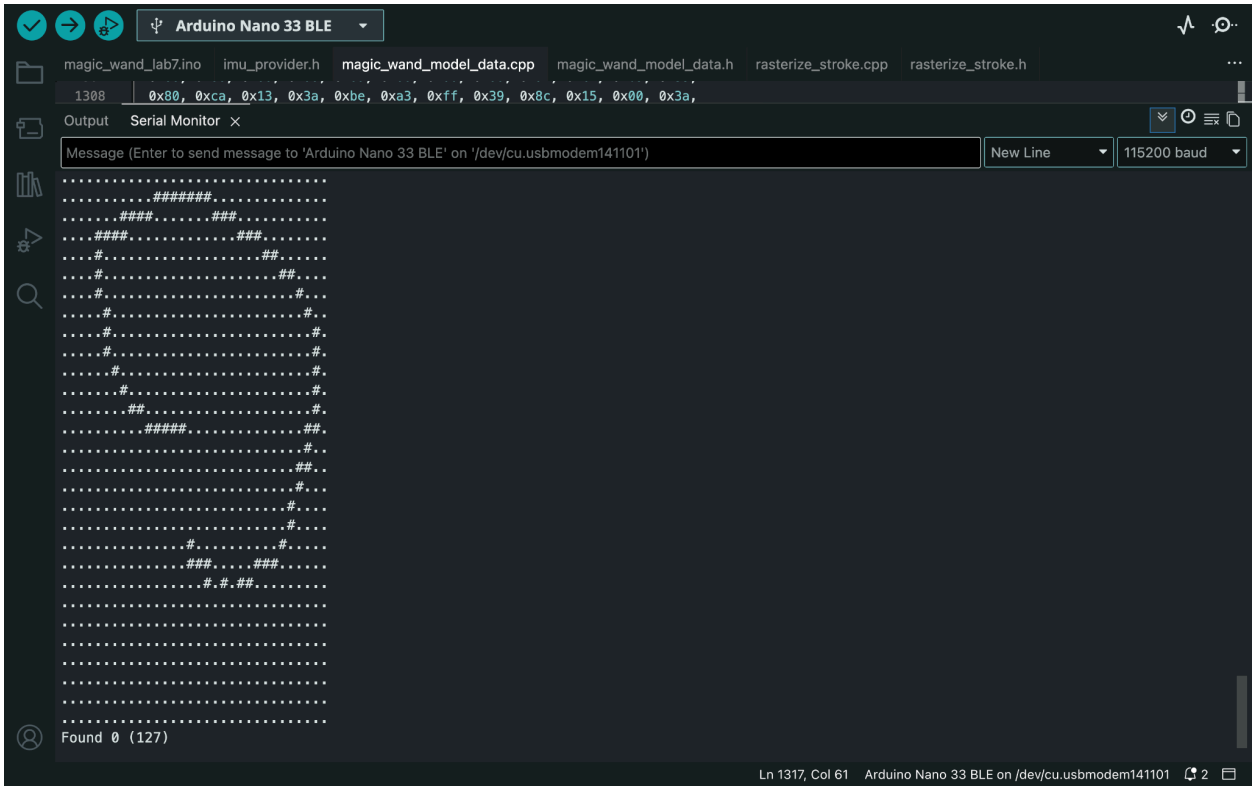
Discussion

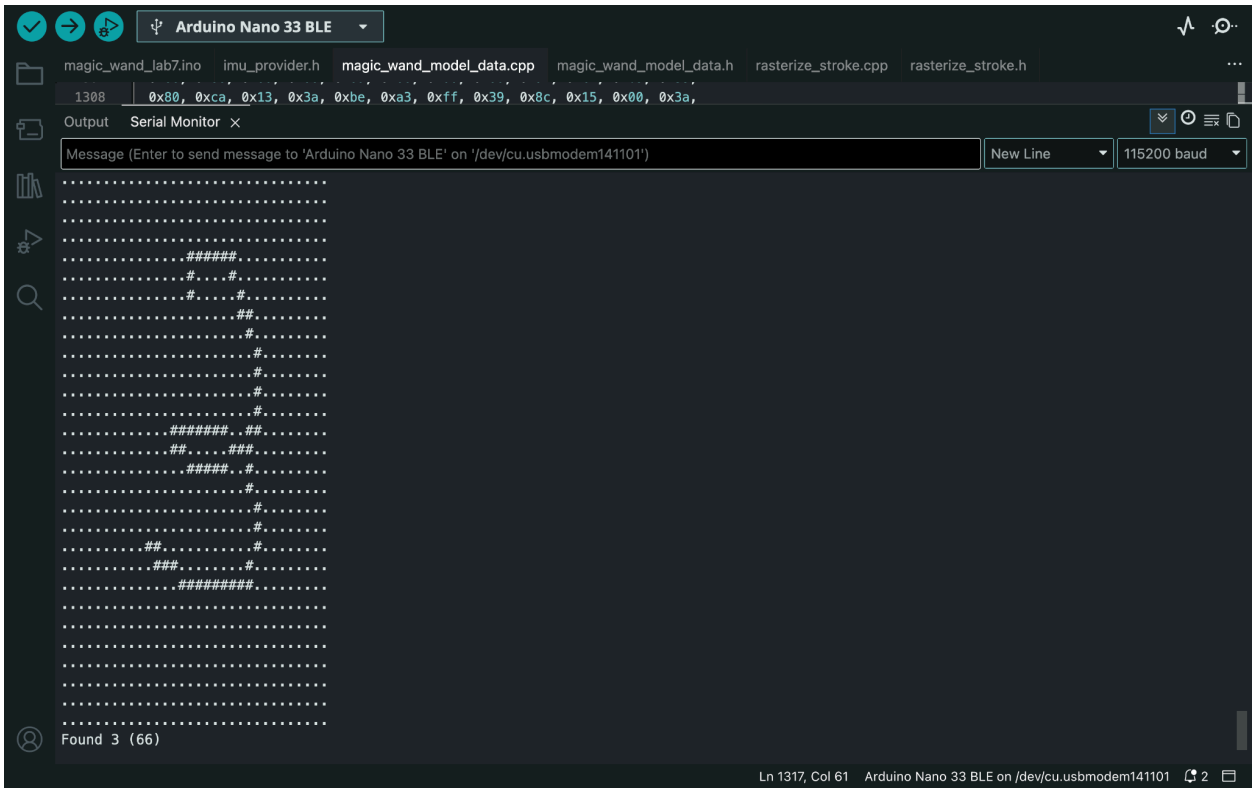
- Between the baseline model and the finetuned model, the baseline model actually performed better than the finetuned in my case. The difference in performance is mainly caused by the poor dataset I collected and used for finetuning.
- For the dataset collection, it was actually split into three person: me, Ben, and Adi. Although we were each supposed to collect 10 labeled numbers from 0-9, the combined dataset ended up with only 74 features, and 19 of those were unlabeled, which means the dataset is really small and skewed.
- Moreover, because the finetuning dataset was collected by three different individuals, that data wasn't tailored to me. Hence during deployment and inference, it wasn't really doing that well for my specific writing habits.
- After identifying this issue, I recollected the data, cleaned the labels, stitched the JSON files together, retrained/finetuned the model, and redeployed it. The new dataset was much more balanced, with 100 total samples and 10 samples per digit class. This led to a clear improvement in the finetuned model. The updated classification report shows an overall accuracy of 93%, with a macro average F1-score of about 0.929. This suggests that the earlier poor result was mainly a data quality problem rather than a limitation of the finetuning method.
- The new model performs especially well on several classes. Digits 1, 2, and 5 all reached perfect precision and recall in the report, meaning the model correctly identified all examples of those digits and did not confuse other digits for them. Digits 3, 4, 7, and 9 also performed strongly, with F1-scores around 0.90–0.95.
- However, the model is still not perfect. The weakest class is digit 8, which has a recall of 0.70 and an F1-score of 0.7778. From the confusion matrix, some 8s were misclassified as 0s and 6s. This makes sense because these digits have visually similar rounded stroke patterns, especially when the wand motion is noisy or inconsistent. There were also smaller confusions such as 0 being classified as 8, 3 as 7, 7 as 0, and 9 as 4. These mistakes show that the model is still sensitive to overlap between similar gesture shapes.
- Overall, the second finetuning attempt was much more successful. The recollected and cleaned dataset improved the model from a poorly generalized finetuned version into one that achieved strong classification performance after redeployment. The main takeaway is that finetuning is highly dependent on dataset quality. A small or mislabeled dataset can make finetuning worse than the baseline, but a clean and balanced dataset can significantly improve the deployed model's performance. Further improvements would likely come from collecting more examples per class, especially for confusing digits like 0, 6, and 8, and collecting more samples under realistic deployment conditions so the model becomes more robust to variation in writing speed, size, and motion style.

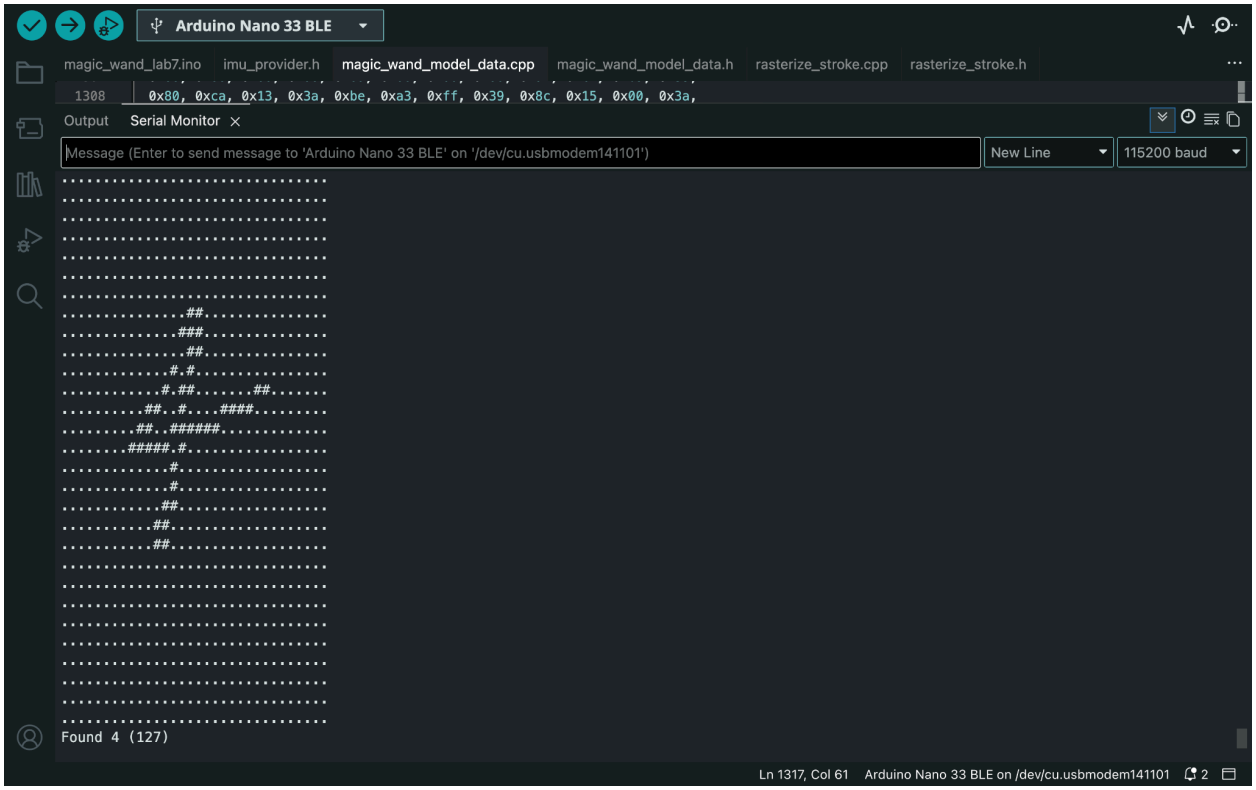
Baseline model

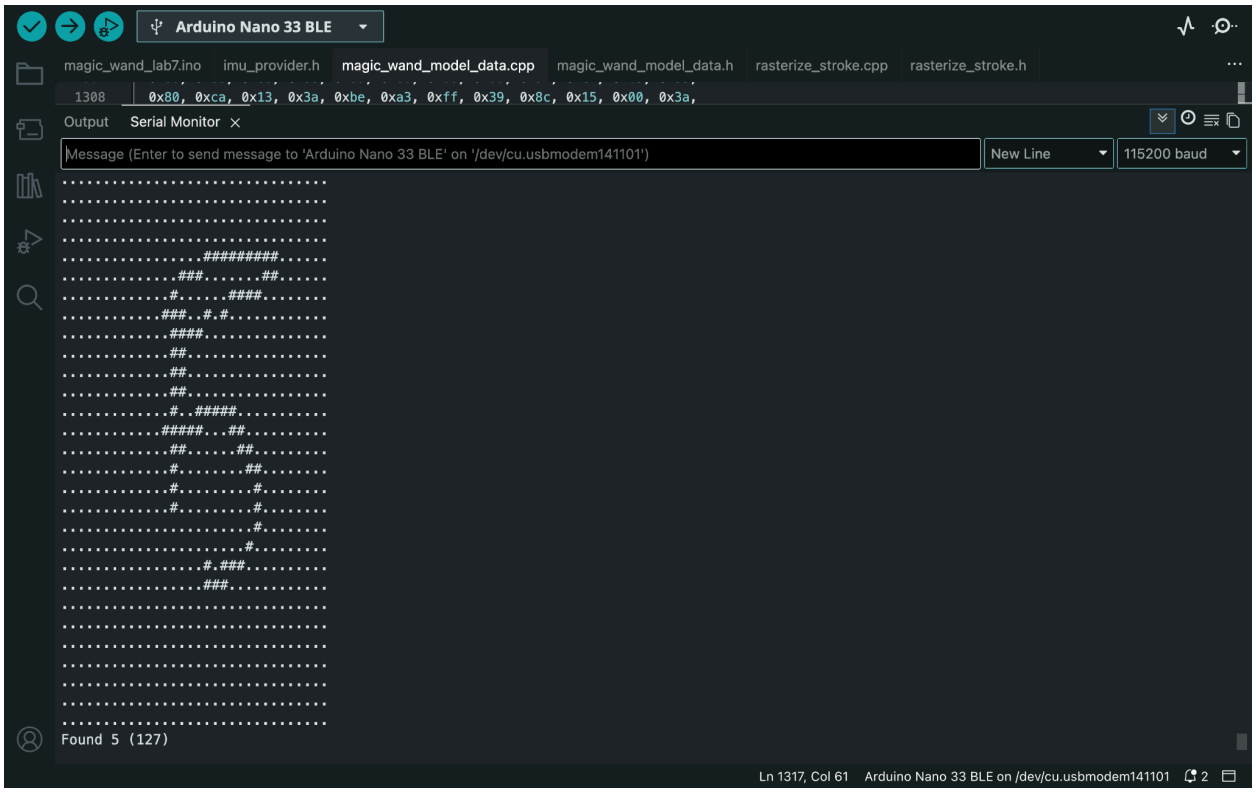


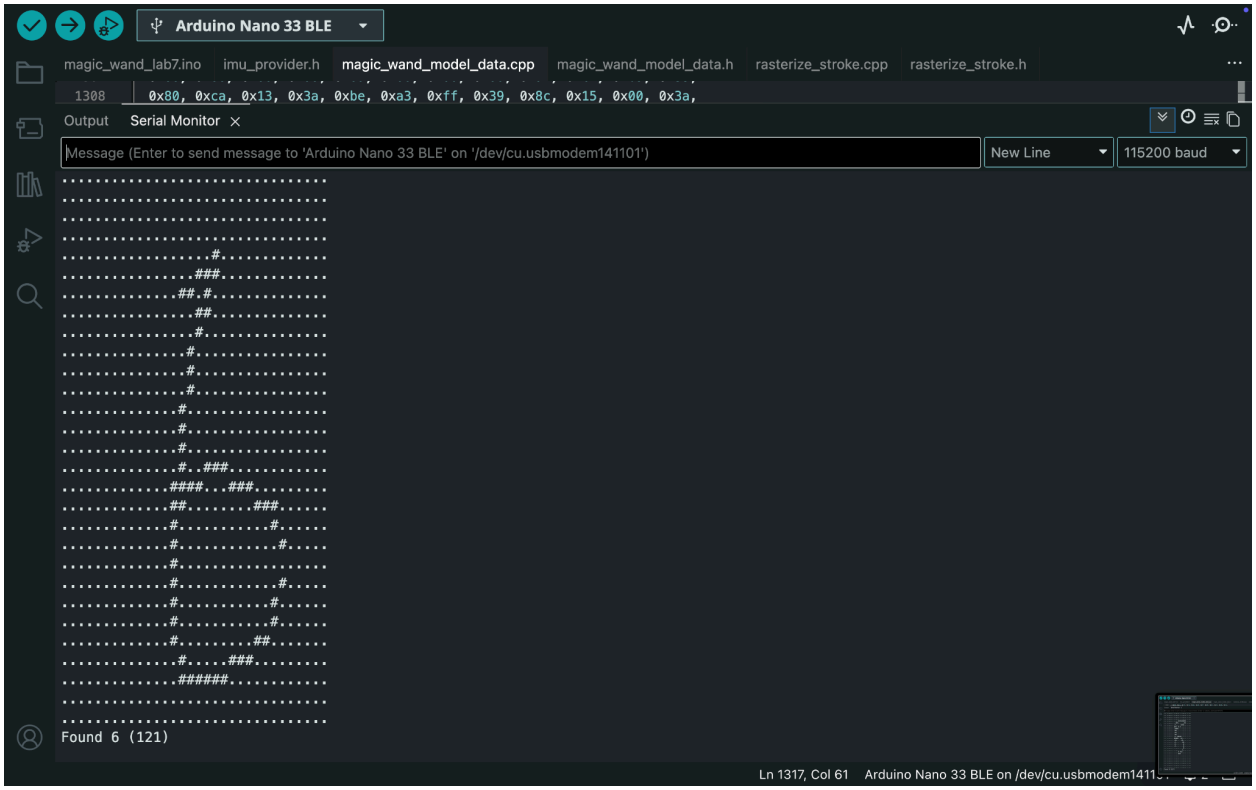


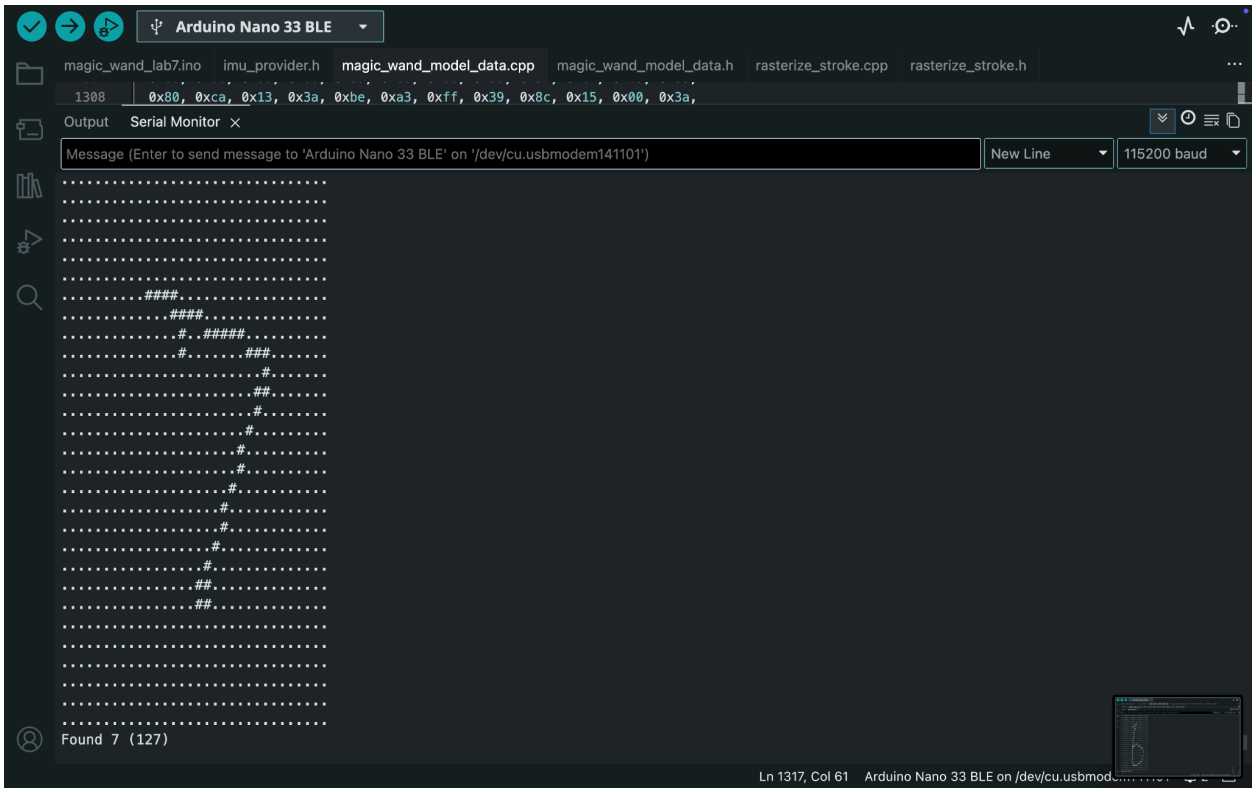

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
1308 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 0 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2

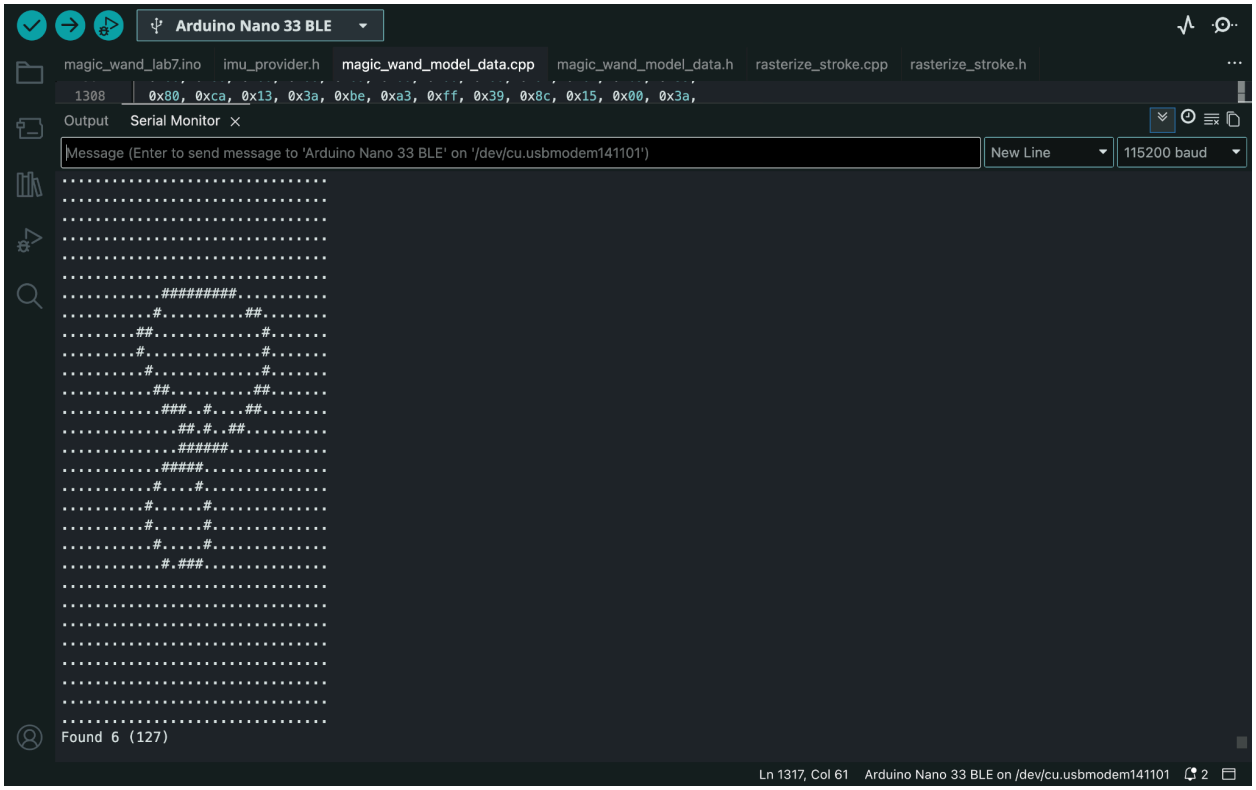

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
1308 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 3 (66)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2

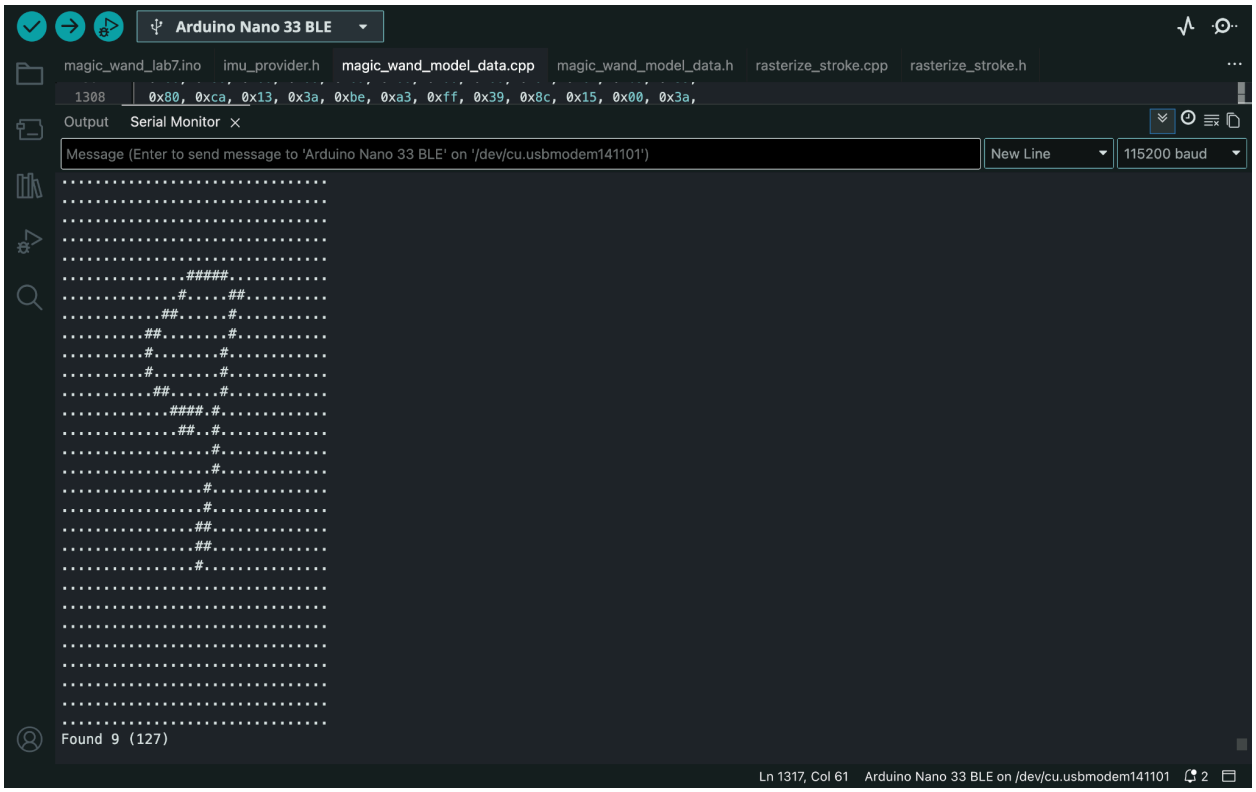

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
1308 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 4 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2


Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
1308 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 5 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2

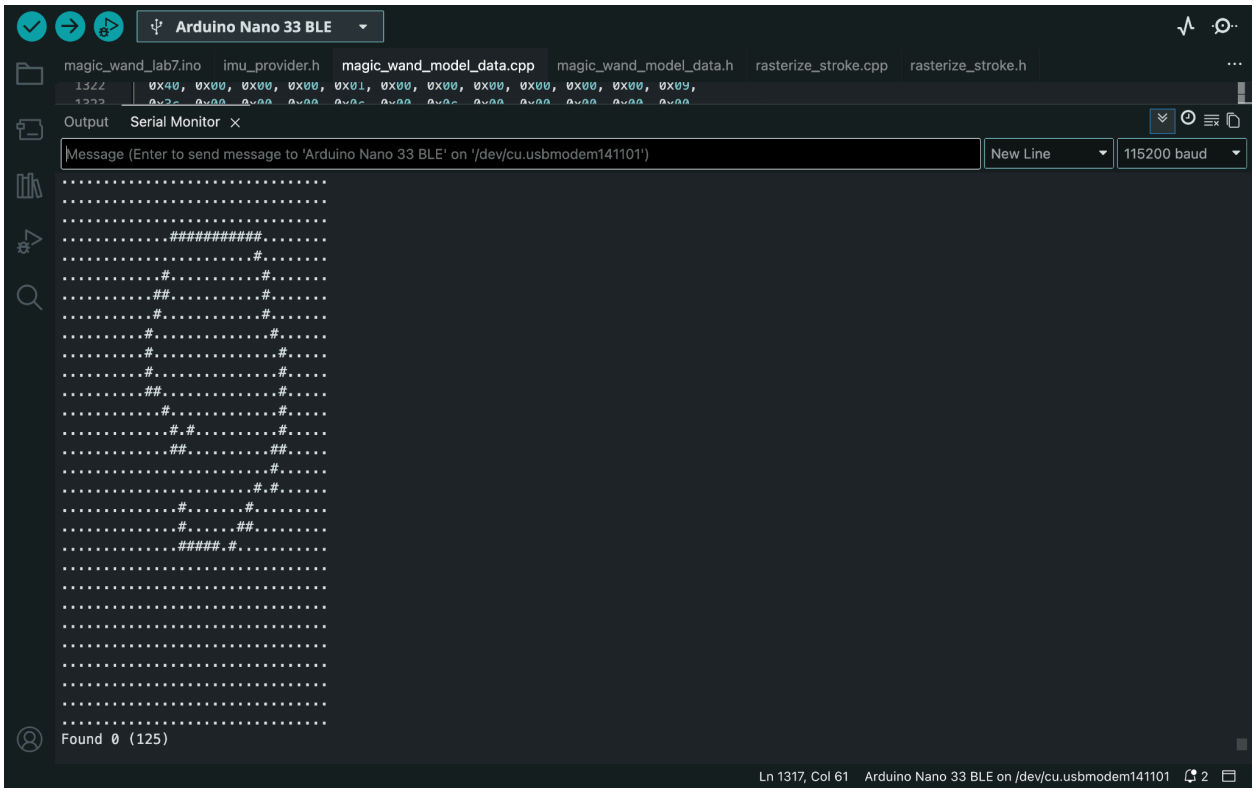

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
1308 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 6 (121)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141

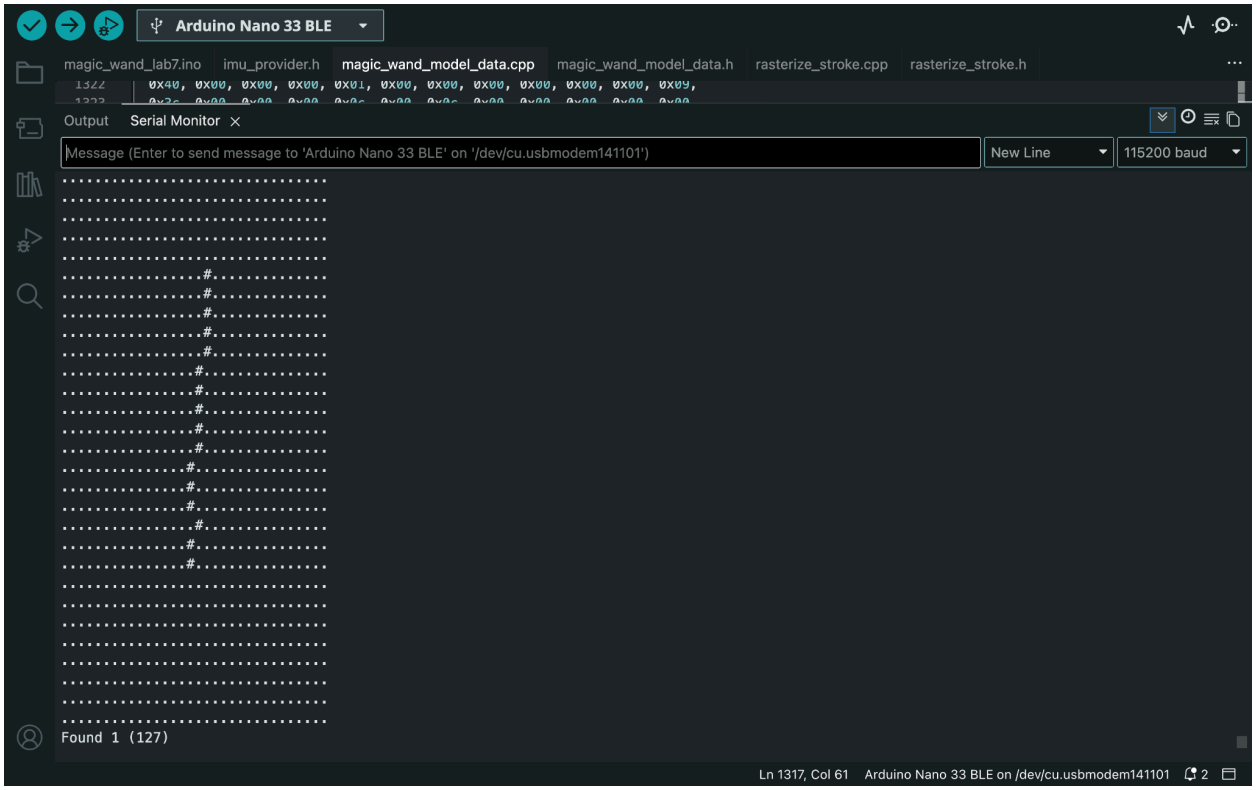

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
1308 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 7 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmod

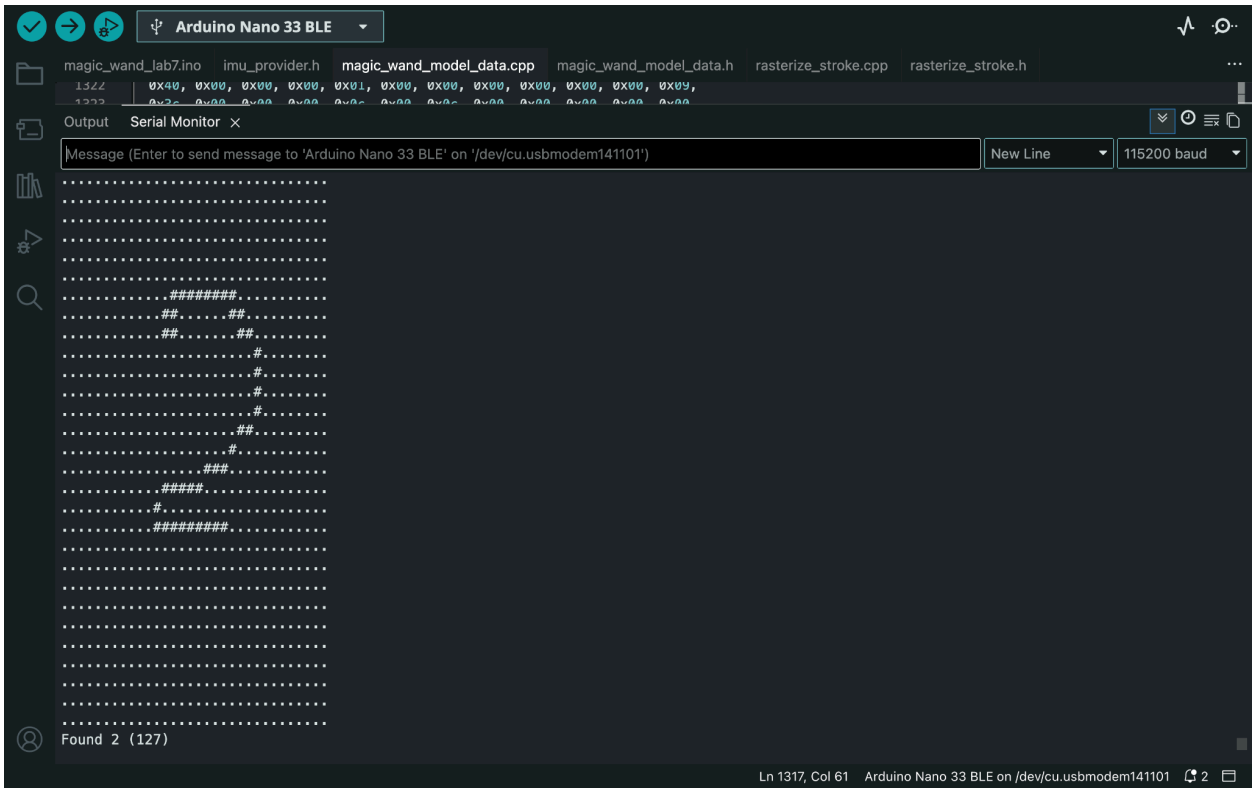

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
1308 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 6 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2

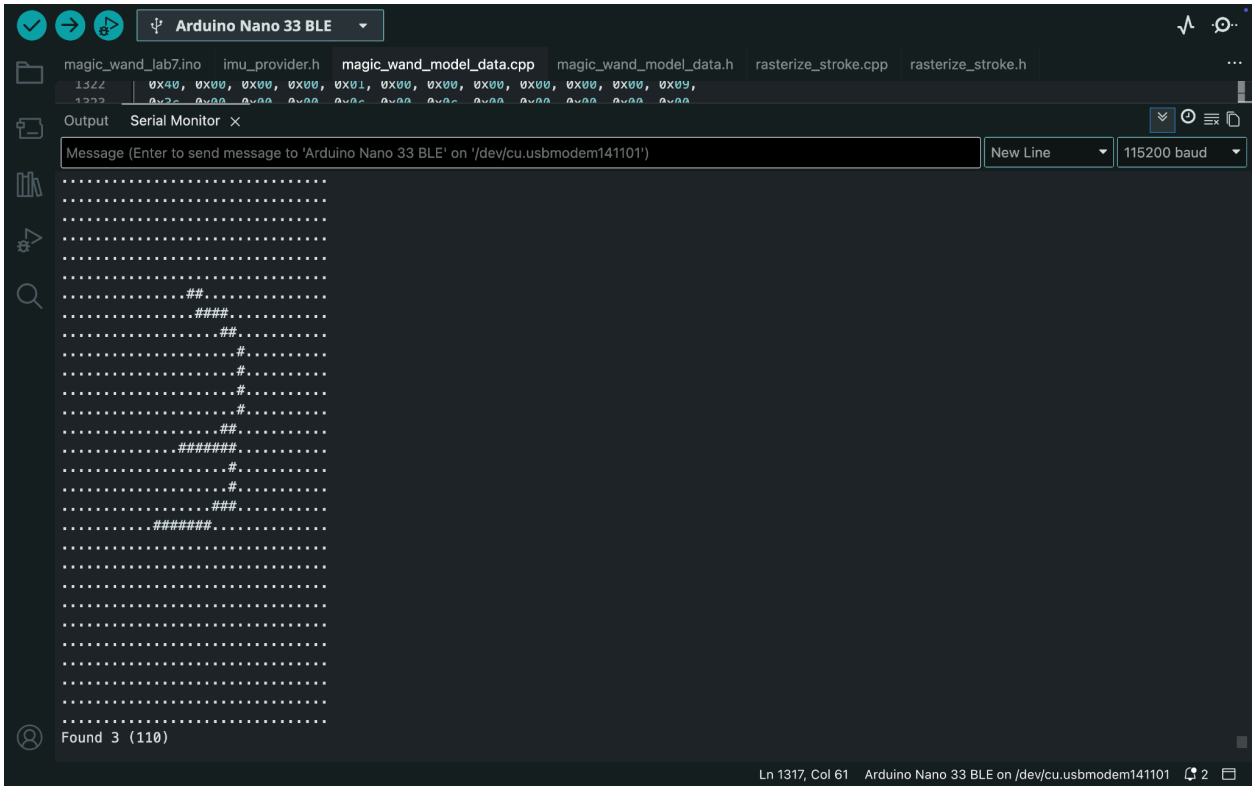

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
1308 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 9 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2

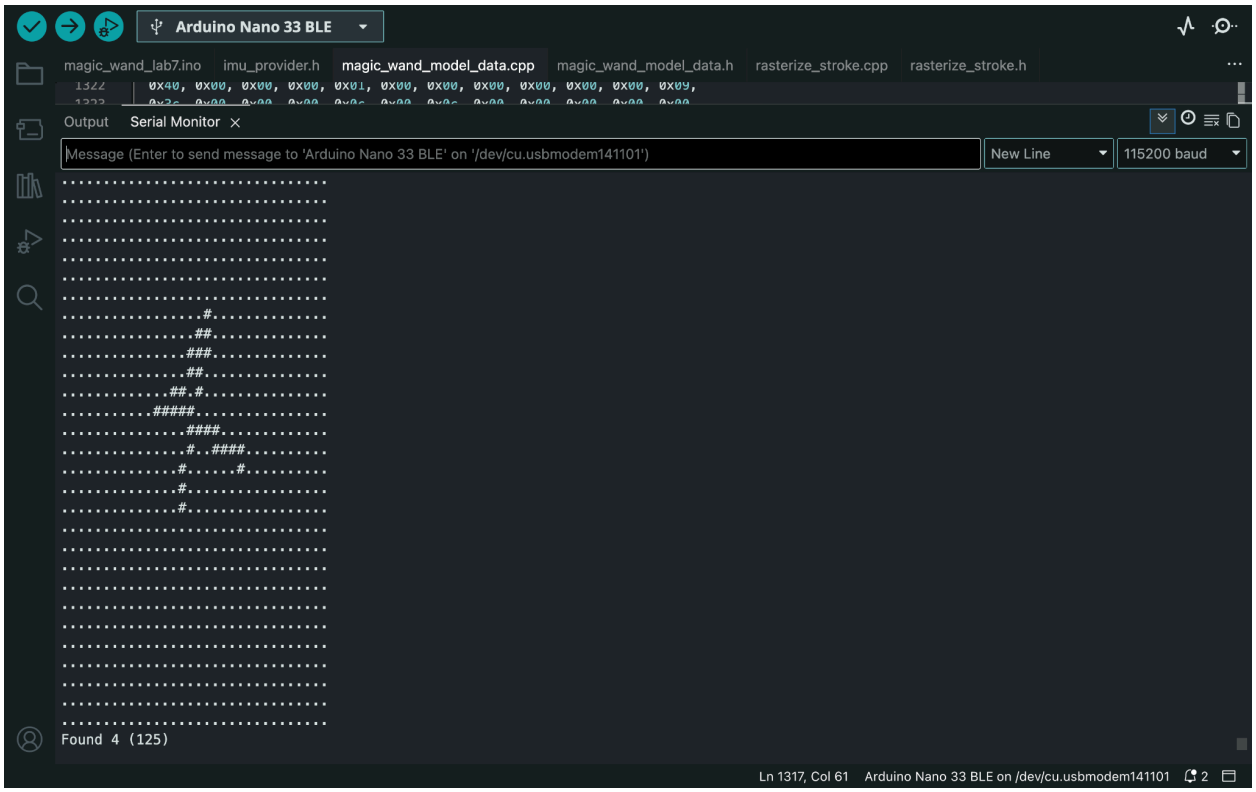
Finetuned model

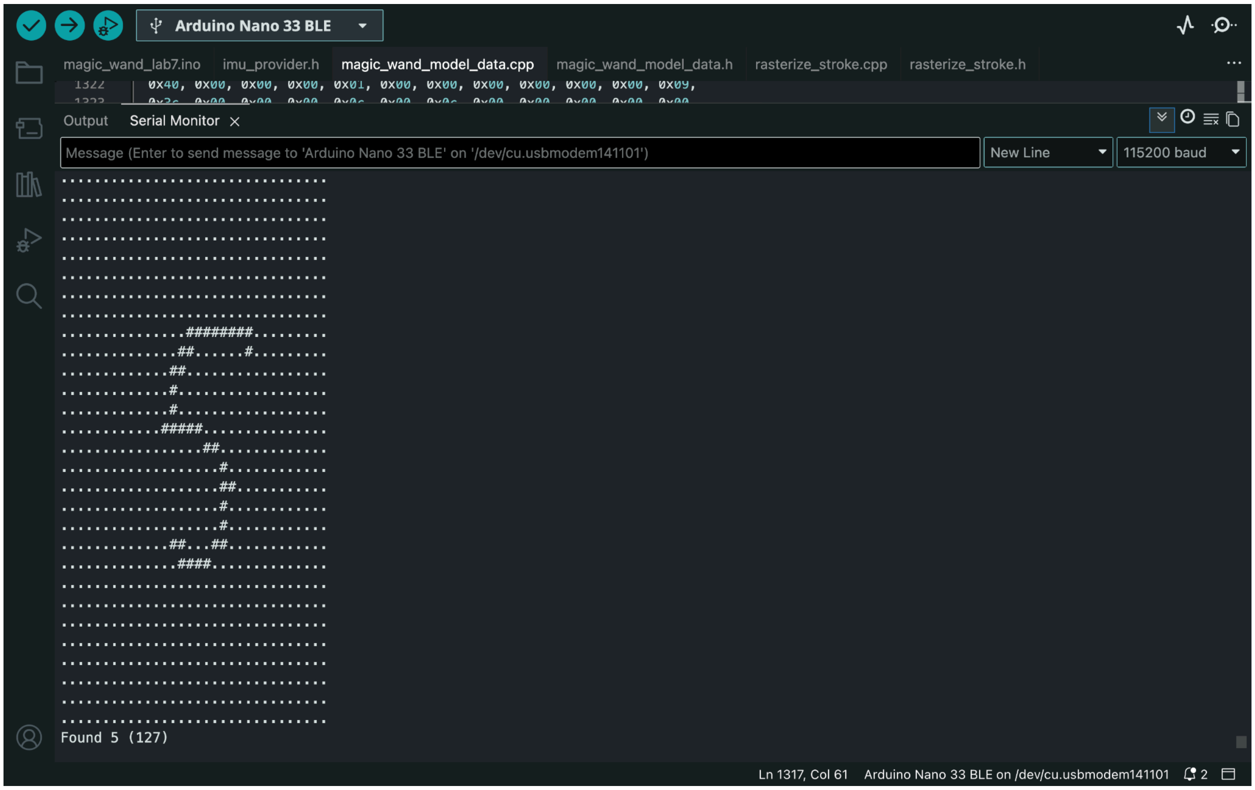


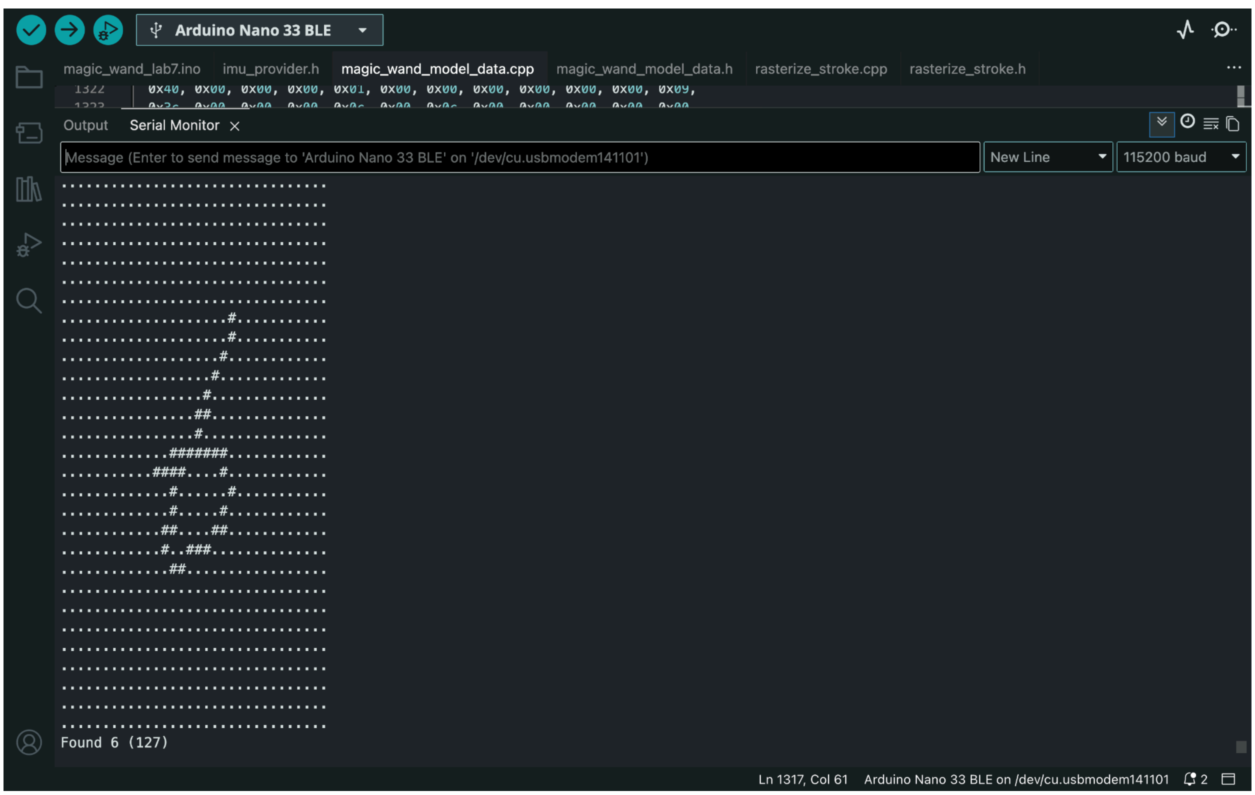

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
0x40, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x09,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 1 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2

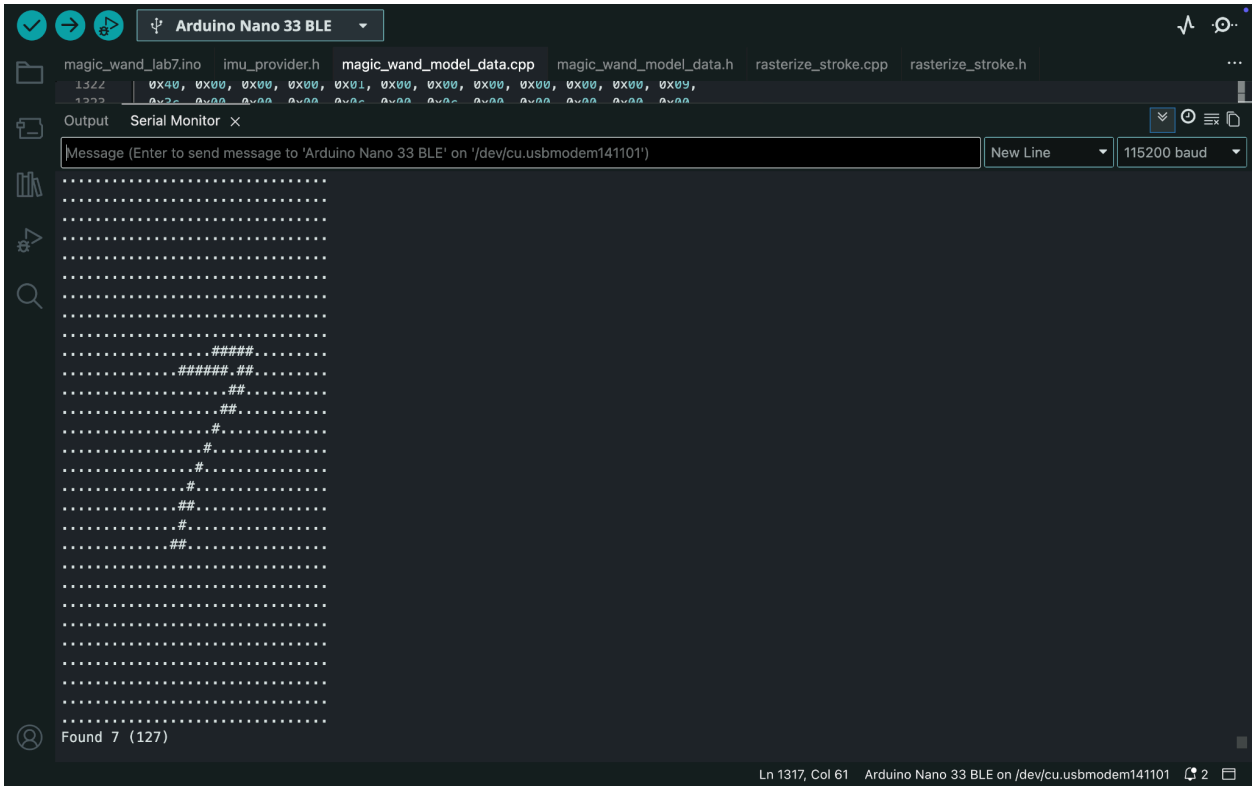

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
0x40, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x09,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 2 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2

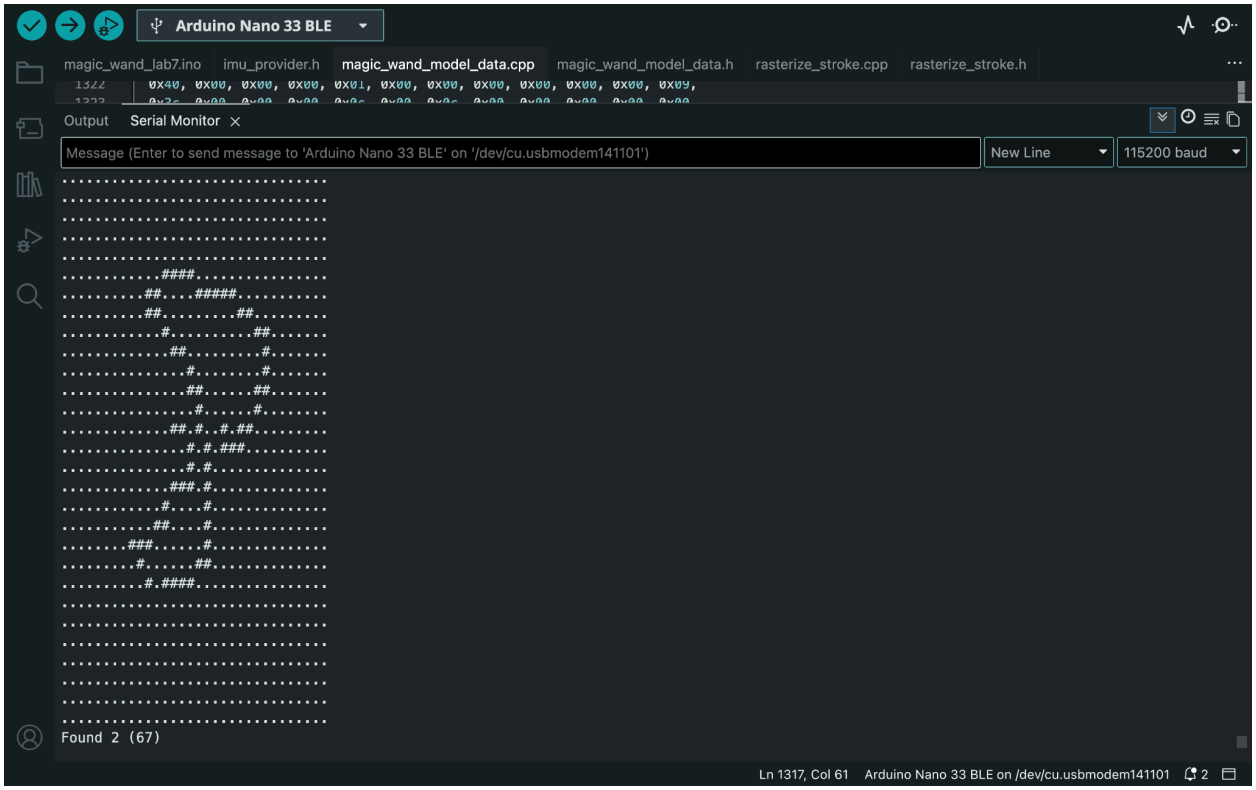

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
0x40, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x09,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 3 (110)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2

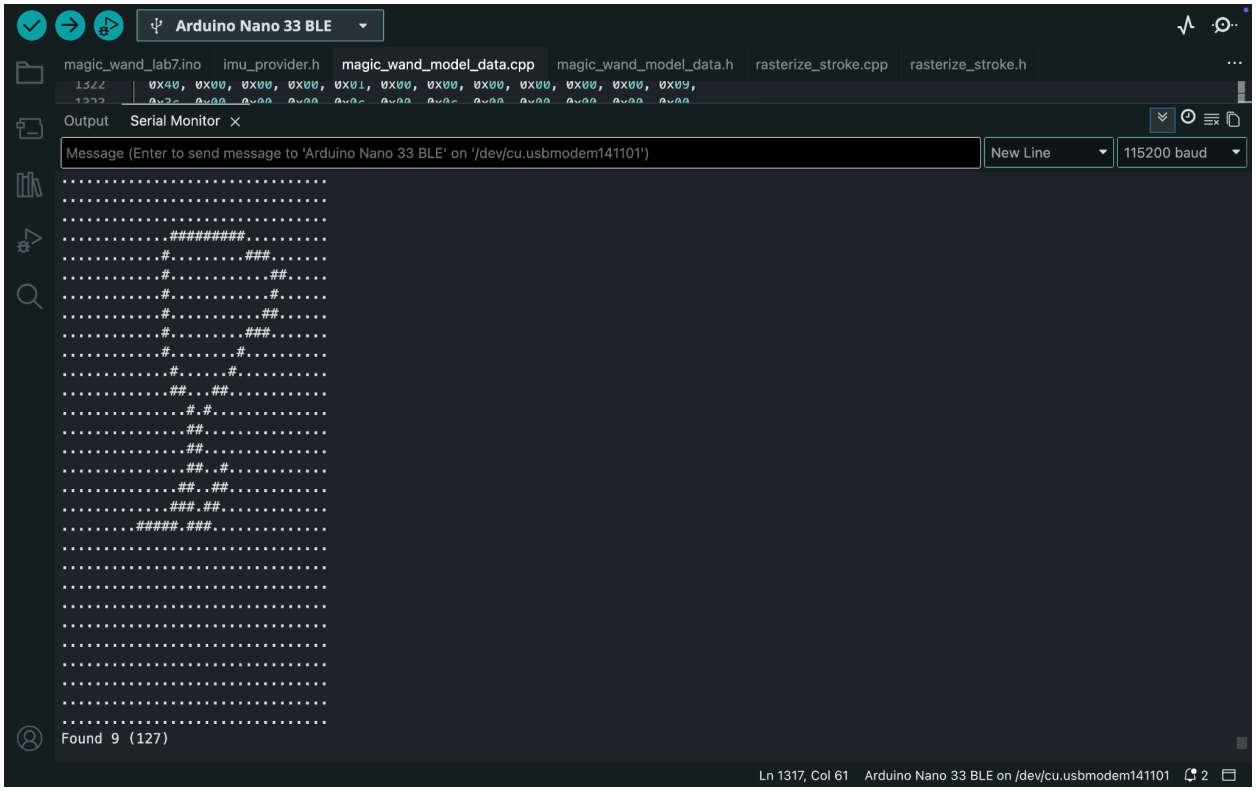

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
0x40, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x09,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 4 (125)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2


Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 5 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101


Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 6 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101


Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
0x40, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x09,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
.....#####.....
.....######.##.....
.....##.....
.....##.....
.....#.....
.....#.....
.....#.....
.....#.....
.....##.....
.....#.....
.....##.....
Found 7 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2


Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
0x40, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x09,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 2 (67)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2


Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
0x40, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x09,
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 9 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101
2